ECE 374 B - Summer 2026

Homework 9

- **Submit your solutions electronically on the course Gradescope site as PDF files.** If you plan to typeset your solutions, please use the LaTeX solution template on the course web site. If you must submit scanned handwritten solutions, please use a black pen on blank white paper and a high-quality scanner app (or an actual scanner, not just a phone camera). We will mark difficult to read solutions as incorrect and move on.
- **Every homework problem must be done *individually*.** Each problem needs to be submitted to Gradescope before midnight of the due data which can be found on the course website: https://ecealgo.com/su26/homeworks.html.
- For nearly every problem, **we have covered all the requisite knowledge required to complete a homework assignment prior to the "assigned" date.** This means that there is no reason not to begin a homework assignment as soon as it is assigned. Starting a problem the night before it is due a recipe for failure.

Policies to keep in mind

- **You may use any source at your disposal**—paper, electronic, or human—but you ***must*** cite ***every*** source that you use, and you ***must*** write everything yourself in your own words. See the academic integrity policies on the course web site for more details.
- **Being able to clearly and concisely explain your solution is a part of the grade you will receive.** Before submitting a solution ask yourself, if you were reading the solution without having seen it before, would you be able to understand it within two minutes? If not, you need to edit. Images and flow-charts are very useful for concisely explain difficult concepts.

See the course web site (https://ecealgo.com/su26) for more information.

If you have any questions about these policies,
please don't hesitate to ask in class, in office hours, or on Piazza.

1. Wuhan's Lab of Virology is attempting to *synthesize* a virus from bat DNA sequences to resemble a dangerous strain. You may not be able to solve everything, but neither were they (like the question of "how did this get out?")

   A bat DNA sequence is a string $b \in \{A, G, T, C\}^*$. The lab defines a computerization function where `ASCII`(c) is the 8-bit ASCII code of the capital letter c, and & is bitwise AND. The function is applied character-by-character, yielding a binary string.

$$\text{computerize}(c) = \text{ASCII}(c) \ \& \ 0xF$$

   We define a searcher which determines whether the computerized version of bat DNA b satisfies the virus construction criterion.

$$\text{VirusSearcher}(x) = \left\{ x \in \{0,1\}^* \;\middle|\; |x| = \left\lceil k\sqrt{k} \right\rceil \text{ for some } k \in \mathbb{N} \right\}$$

$$L_{\text{search}}(b) = \text{VirusSearcher}(\text{computerize}(b))$$

   The lab also develops a set of Turing machines $D_1, \ldots, D_x$, each acting as a *virus detector*. A detector D_i is said to **detect** a sequence $w \in \{A, G, T, C\}^*$ if and only if $D_i(w)$ halts and accepts.

   Answer the following:

   a. Is VirusSearcher **decidable** or **undecidable**? Justify your answer.

   b. Is VirusSearcher **regular** or **irregular**?.

   c. Is the problem of deciding whether this *specific* language VirusSearcher, the one defined in the question, **decidable** or **undecidable**? Prove it!

   d. Prove that L_{search} is **regular**, if VirusSearcher is regular.

   e. Three days later, the lab director falls ill with COVID and, in a feverish act of inspiration, obfuscates every virus detector by applying a transformation f such that:

      - f is not invertible,
      - For every M, $L(f(M)) = L(M)$,
      - $f(\langle M \rangle)$ does not contain any recognizable substrings from the original.

      Given the transformed machines $f(D_1), \ldots, f(D_x)$ and the original virus strings $B_1, \ldots, B_m$, is it **decidable** or **undecidable** to determine whether any detector accepts any virus string as input? Prove it.

2. **(The Final Mutation)** The same lab releases a new generation of *bounded* detectors. Each detector D_i now halts within a known polynomial bound $t(n)$ on inputs of length n. Given the obfuscated and bounded detectors $f(D_1), \ldots, f(D_x)$, virus strings $B_1, \ldots, B_m$, and a threshold $k \in \mathbb{N}$, consider the following question:

   That is, does there exist a subset $S \subseteq \{1, \ldots, x\}$ with $|S| \geq k$ such that for every $i \in S$, there exists a j for which $f(D_i(B_j))$ halts and accepts within the step bound?

3. Suppose k is polynomial in the input size. Is this problem:

   Decidable, Undecidable, P, NP NP-hard? NP-complete.

4. Now suppose k is arbitrary. No formal proof needed. Justify, even an intuitive explanation is fine.

   Decidable, Undecidable, P, NP, NP-hard, NP-hard?